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Trends in Initiation of Buprenorphine for Opioid Use Disorder Among Older Adults

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INTRODUCTION

In the last decade, the prevalence of opioid use disorder (OUD) in US adults aged 65 years and older has tripled.¹ In January 2020, access to buprenorphine, one of three medications for opioid use disorder (MOUD), improved through expansion of Medicare bundled

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payments for office-based and telehealth-based OUD treatment and opioid treatment programs.^{2,3} Additional policy efforts expanded under Medicaid coverage requirements for buprenorphine beginning in October 2020.⁴ Whether efforts to expand access to OUD care have impacted MOUD initiation patterns in older adults is unknown.⁵ Thus, we examined national trends in buprenorphine initiation in adults 65 and older across payers between 2018 and 2021.

METHODS

We analyzed buprenorphine prescriptions dispensed to adults aged 65 years and older between January 2018 through June 2021 using the IQVIA Longitudinal Prescription Database which covers 92% of dispensations in US retail pharmacies. We included buprenorphine prescriptions in sublingual film or tablet form and excluded formulations for pain management and injectable prescriptions. We linked national provider identifiers (NPI) from IQVIA to National Plan and Provider Enumeration System and National Uniform Claim Committee data to identify prescriber specialty. We extracted payer from prescriptions and categorized into Medicare, Commercial, Medicaid, Cash or Coupon, and Other (Supplemental Methods).

We defined initiation as a first buprenorphine prescription without prior fills in the preceding 180 days.^{6,7} We calculated buprenorphine initiation rate as the number of new users per 100,000 older adults based on population estimates from the U.S. Census. To identify monthly percent change in buprenorphine initiation rate, we performed joinpoint regressions with significance level $p < 0.05$ (Supplemental Methods). Analyses were performed using Joinpoint version 5.4.0.0 and R Statistical Software v4.5.0.^{8,9}

RESULTS

During the study period, 86,704 older adults initiated buprenorphine (median [IQR] age 69 [66–74], 47.1% female) (Table 1). The most common payers were Medicare (55.1%) and Cash or Coupon (16.5%). The most common specialty of buprenorphine prescribers was primary care clinicians (37.1%), advanced practice clinicians (18.4%), and anesthesia pain medicine clinicians (9.6%). Geriatricians accounted for less than 1% of buprenorphine initiations.

Between January 2018 and March 2019, the monthly buprenorphine initiation rate increased from 3.0 to 4.1 per 100,000 older adults, with an estimated monthly percentage increase of 2.2% (95% CI, 0.6% to 3.9%; $p = 0.01$) (Figure 1A). Between March 2019 and June 2021, the monthly buprenorphine initiation rate was stable (monthly percentage change 0.3%; 95% CI, -0.2% to 0.8%; $p = 0.25$).

Temporal trends varied by payer (Figure 1B). Between January 2018 and April 2019, the monthly initiation rate for Medicare payers increased by 3.2% (95% CI, 1.8% to 8.8%; $p < .001$), with no significant changes between April 2019 to June 2021. The monthly initiation rate for Medicaid payers steadily increased by 0.7% (95% CI, 0.2% to 1.2%; $p = 0.01$) throughout the entire study period. In contrast, the monthly initiation rate for Cash

or Coupon payers steadily decreased by 0.5% (95% CI, -2.4% to -0.1%; $p=0.03$) between January 2018 and November 2020.

DISCUSSION

Monthly buprenorphine initiation rates in older adults increased between 2018 and early 2019 and then plateaued from April 2019 through June 2021. The plateauing of buprenorphine initiation indicates that neither Medicare efforts to increase access to MOUD through bundled payments for office-based OUD care nor Medicaid coverage expansion increased buprenorphine initiation in older adults, which could in part be due to other contemporaneous events such as the COVID-19 pandemic.

These findings build upon a recent cross-sectional study of buprenorphine prescribers to older adults, also using IQVIA data, which found a small subset of clinicians account for most buprenorphine prescriptions to older adults, few of whom were geriatricians.⁵ As the population with OUD ages, there may be a role for both expanded buprenorphine prescribing by geriatricians and expansion of geriatric principles to other prescribers managing older adults with OUD.

Study limitations include lack of data on race or ethnicity and data on buprenorphine directly administered through opioid treatment programs rather than prescribed. We did not examine older adults dually eligible for Medicare and Medicaid given data limitations. Our study period ended June 2021, thus we are not able to examine more recent trends in buprenorphine prescribing, including the removal of the “x-waiver” in 2023, which further reduced access barriers.¹⁰

As the rate of OUD is growing among U.S. older adults, these findings suggest additional implementation efforts will be needed in addition to policy changes to increase uptake of buprenorphine in older adult populations. These efforts could include initiatives to enhance screening for OUD, particularly among older adults on long-term opioid therapy, and clinician education on prescribing MOUD in geriatric and other primary care settings.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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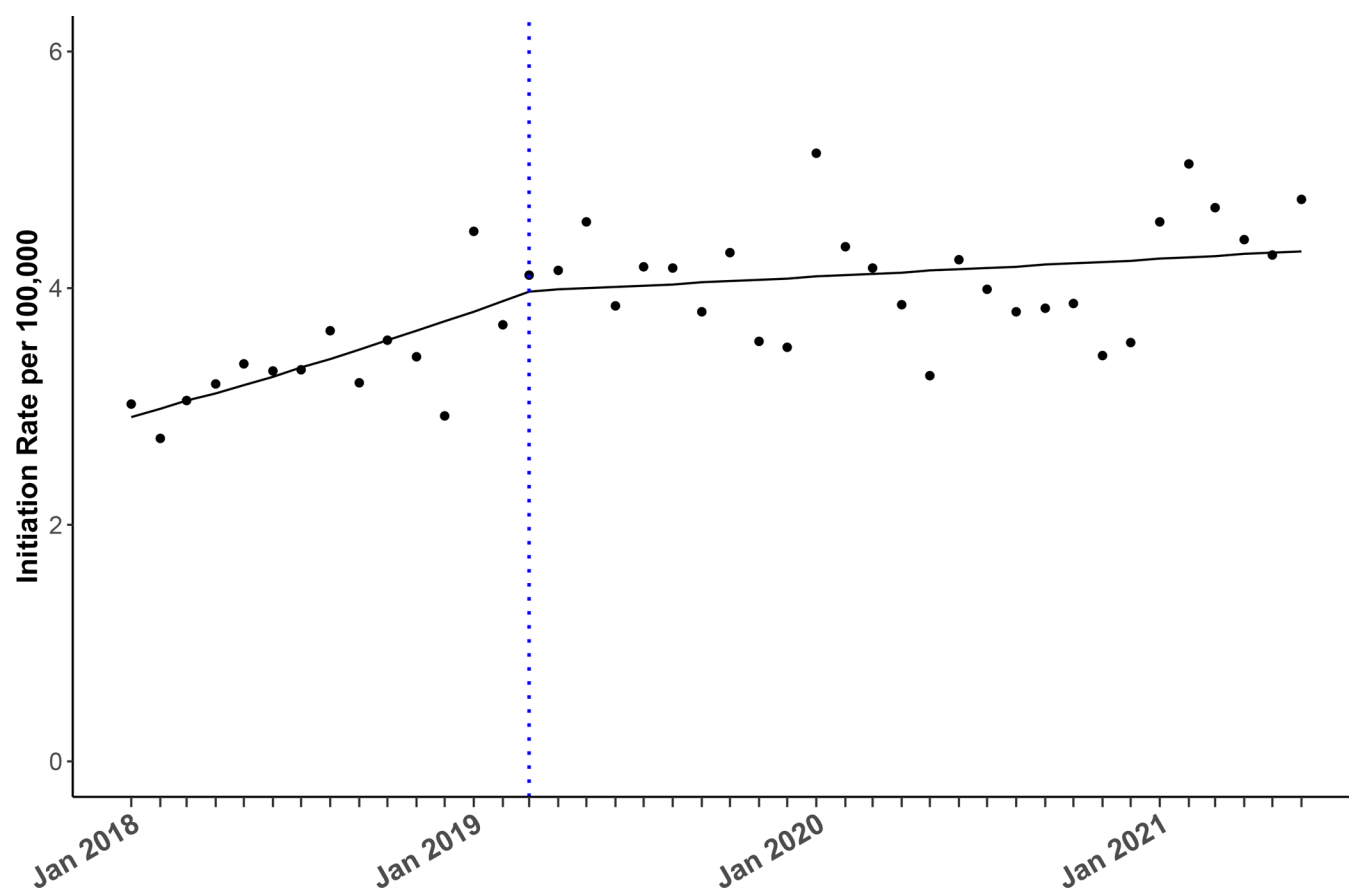
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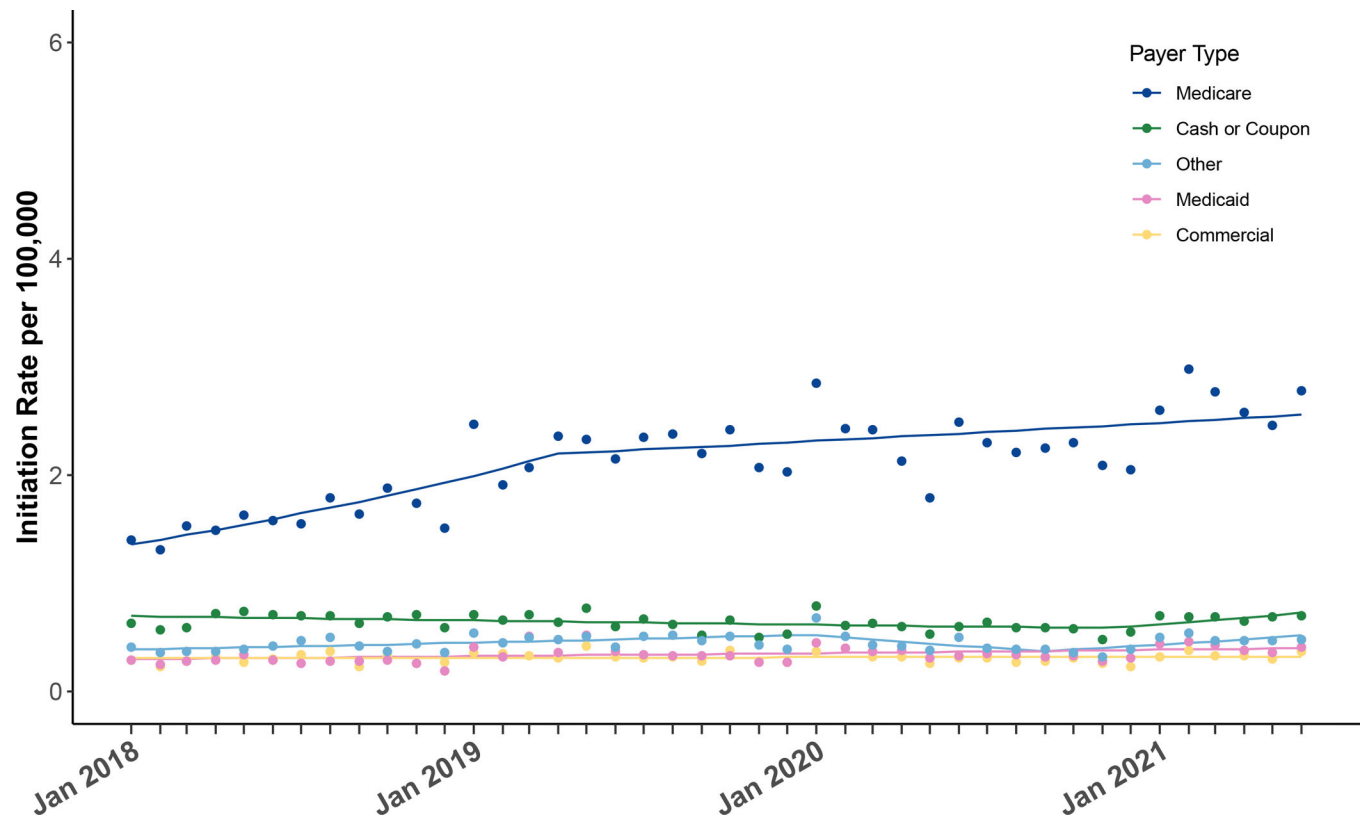
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Panel A: The vertical blue line represents significant slope change identified by joinpoint regression.

Segment 1 Range: MPC (95% CI): 2.2% (0.6% to 3.9%). Segment 2 Range: MPC (95% CI): 0.3% (-0.2% to 0.8%).



Panel B: Medicare Joinpoint: Segment 1 Range: MPC (95% CI): 3.2% (1.9% to 8.8%), Segment 2 Range: MPC (95% CI): 0.6% (-0.5% to 1.0%). Cash or Coupon Joinpoint: Segment 1 Range: MPC (95% CI): -0.5% (-2.4% to -0.1%). Segment 2 Range: MPC (95% CI): 3.1% (-0.8% to 15.4%). Other Joinpoint: Segment 1 Range: MPC (95% CI): 1.2% (0.4% to 2.7%), Segment 2 Range: MPC (95% CI): -4.1% (-14.0% to -0.5%), Segment 3 Range: MPC (95% CI): 3.7% (0.3% to 12.2%). Medicaid Joinpoint: Segment 1 Range: MPC (95% CI): 0.7% (0.2% to 1.2%). Commercial Joinpoint: Segment 1 Range: MPC (95% CI): 0.1% (-3.0% to 0.6%).

Table 1.
 Characteristics of Older Adults Initiating Buprenorphine, 2018–2021

Characteristic	No. (%) (N= 86,704)
Age Categories, years	
65 to 69	45,489 (52.5)
70 to 74	19,972 (23.0)
75 to 79	9,173 (10.6)
80 and older	12,070 (13.9)
Female	40,833 (47.1)
Payer Type	
Medicare	47,793 (55.1)
Cash or Coupon	14,321 (16.5)
Other	9,977 (11.5)
Medicaid	7,653 (8.8)
Commercial	6,960 (8.0)
Prescribing Provider Specialty	
Primary Care	32,139 (37.1)
Advanced Practice Clinicians	15,931 (18.4)
Anesthesia / Pain Medicine	8,303 (9.6)
Psychiatry & Neurology	8,035 (9.3)
Surgical Specialty	5,143 (5.9)
Addiction Medicine	4,974 (5.7)
Other	3,457 (4.0)
Physical Medicine & Rehabilitation	2,241 (2.6)
Addiction Psychiatry	1,173 (1.4)
Medical Specialty	1,039 (1.2)
Other Physician	911 (1.1)
Geriatrics/Gerontology	665 (0.8)
Missing	2,693 (3.1)